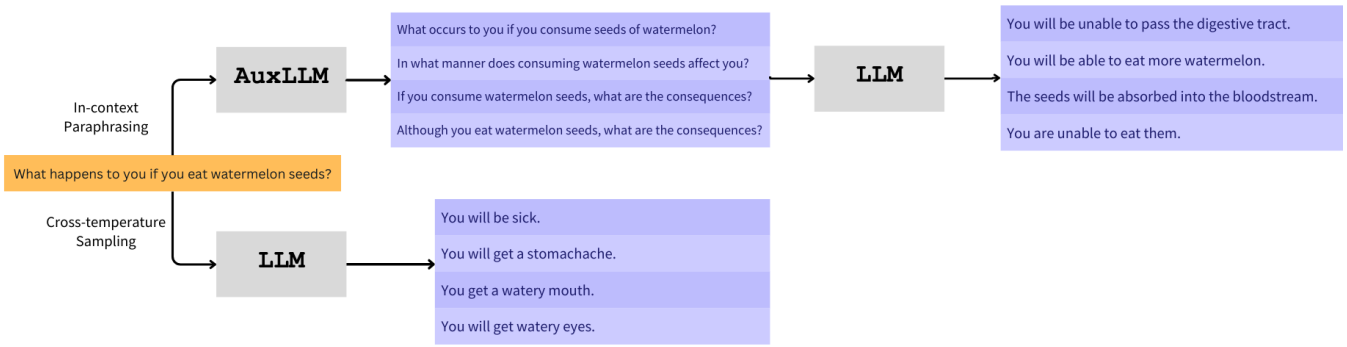

Figure 2: Examples of the two methods of paraphrasing and answer generation.

Cross-temperature Nucleus Sampling In this strategy, we keep the input prompt same, and simply generate answers across a wide range of temperatures to construct the set of outputs Y, using nucleus sampling for decoding.

As a concrete example, we look at Fig. 2. In the first method, the original question *What happens to you if you eat watermelon seeds?* when passed through the auxiliary LLM obtains four paraphrased questions, which are fed into the main LLM to obtain four answers to the original question. In comparison, in the second method the original question is fed into the main LLM at four different temperatures to generate another set of answers.

Improving Consistency by Asking to Choose

Pretrained language models show marked improvements in complex task-specific performances when they are finetuned on a range of relevant instructions, and/or using RLHF (Lambart et al. 2023). Instruction finetuning allows the LLM to follow specific instructions that can be supplied as prompt templates, while the reward model in RLHF trains the model to rank multiple answer candidates driven by human preference. Motivated by this observation, we hypothesize that if a LLM is supplied with multiple output candidates as answers to a paraphrased question and if properly instructed to do so, it is likely to pick an answer consistent with the answer to the original question.

Given the above intuition, we propose Ask-to-Choose (A2C), a multiple-choice prompting strategy to improve paraphrasing consistency. At a high level, we start with sets of output variations across paraphrasing contexts $C = \{c_1, \ldots, c_I\}$ and temperatures $T = \{t_1, \ldots, t_J\}$. Denote these outputs by Y_c and Y_t, respectively. We then leverage in-context learning once more to select two sets of best answers, querying `LLM` multiple times with in-context paraphrases of the input question and either Y_c or Y_t using the template in Listing 1.

We present the details of the above process in Algorithm 1. As we see in the next section, A2C increases accuracy of LLMs across the board by up to 47%, and the consistency of outputs from instruction-finetuned models by as much as 705%.

Algorithm 1: The Ask to Choose Method

Input: Original question q
Parameter: Paraphrasing contexts C, temperatures T.
Output:

// Generate in-context outputs
for $c_i \in C$ **do**
  $q(c_i) \leftarrow \texttt{AuxLLM}(\texttt{paraphraseprompt}(q, c_i))$
  $y(c_i) \leftarrow \texttt{LLM}(\texttt{answerPrompt}(q(c_i)))$
end for
Let $Y_c = \{y(c_i); c_i \in C\}$

// Generate cross-temperature outputs
for $t_j \in T$ **do**
  $y(t_j) \leftarrow \texttt{LLM}(\texttt{answerPrompt}(q, t_j))$
end for
Let $Y_t = \{y(t_j); t_j \in T\}$

// Ask-to-Choose
for $c_i \in C$ **do**
  $\tilde{q}(c_i) \leftarrow \texttt{LLM}(\texttt{paraphraseprompt}(q, c_i))$
  $y_c(c_i) \leftarrow \texttt{LLM}(\texttt{rankPrompt}(\tilde{q}(c_i), Y_c))$
end for
for $t_j \in T$ **do**
  $y_t(t_j) \leftarrow \texttt{LLM}(\texttt{rankPrompt}(\tilde{q}, Y_t))$
end for
Let $\hat{Y}_c = \{\hat{y}_c(c_i); c_i \in C\}$
Let $\hat{Y}_t = \{\hat{y}_t(t_j); t_j \in T\}$

return $\hat{Y}_c, \hat{Y}_t$.

Experiments

Setup

Data To evaluate semantic consistency, we use answers generated on questions from the TruthfulQA benchmark dataset (Lin, Hilton, and Evans 2022), using the two variations mentioned above. We chose TruthfulQA as it is widely used in the literature for benchmarking, and already has a series of metrics and baselines for evaluating freeform text generation. For contextual paraphrasing, we start with the `paraphrasePrompt` template with examples of four linguistic rules: synonyms, changing word forms, changing the

Listing 1: The `rankPrompt` Template for A2C

```
Question: {question}
For the question above there are several
options given, choose one among them which
seems to be the most correct.
Option {1}: {answer1}
Option {2}: {answer2}
Option {3}: {answer3}
Option {4}: Don't know the correct answer
Answer:
```

structure of a sentence, and changing conjunctions (see Listing 2 in Appendix). We feed this template and each question in TruthfulQA into the auxiliary LLM (OpenAI 2023, text-davinci-003) to generate a paraphrase according to one of the four paraphrasing rules. Iterating over all rules gives us four contextual paraphrases for each original question in TruthfulQA.

Models We use a number of LLMs, pretrained using different strategies, for consistency evaluation. Firstly, we use a series of OPT models (Zhang et al. 2022) with increasing number of parameters (from 125M to 6.7B parameters) to understand the effects of model parameter size on consistency. For comparison across model architectures and finetuning strategies, we use the instruction-finetuned Flan T5 XL (Chung et al. 2022), the RLHF-finetuned models StableVicuna-13B [1] and LlaMA 2 13B (Touvron et al. 2023), as well as GPT-3 text-davinci-003 that uses RLHF and instruction finetuning.

Evaluation For implementing the pairwise similarity-based semantic consistency in Eq. (3), we use four measures of semantic equivalence as $s(\cdot, \cdot)$:

1. Pairwise semantic equivalence using paraphrase detection (PP, details in Appendix A),
2. Pairwise agreement or entailment (Entail), and
3. Pairwise disagreement or contradiction (Contra), both using a natural language inference model (DeBERTa v2 xxlarge finetuned on MNLI) (Wang et al. 2018),
4. Semantic cluster entropy (Eq. 4).

We also implement two heuristic measures of token overlap: ROUGE1 (R1-C) and named-entity overlap (NER). Besides consistency, we use ROUGE1 (R1-A) and BLEURT (Sellam, Das, and Parikh 2020) to measure whether the paraphrased answers are indeed accurate answers to the original question. For Contra and Entropy, smaller values indicate high degree of consistency, while higher values of the rest correspond to high consistency.

Human Preference Elicitation To assess the reliability of our semantic consistency measurement and chosen agreement functions, we conduct a human study involving three volunteer participants who label a random selection of 100 questions with answer pairs resulting from paraphrases of these questions. Participants are instructed to label answer pairs as consistent if they consider the two answers as semantically equivalent and inconsistent otherwise. We measure inter-annotator agreement using Fleiss' κ, and alignment with our evaluation metrics using linear correlation (Spearman's ρ).

[1] https://huggingface.co/spaces/CarperAI/StableVicuna

Model	Context		Temperature	
	PP	PP | Acc	PP	PP | Acc
OPT-125M	4.1	1.1	8.3	8
OPT-350M	2.1	1.2	6.5	7.3
OPT-1.3B	22.4	18.3	35.9	30.2
OPT-6.7B	24.4	19.7	32.5	27.8
Flan-T5 XL	43.6	36.5	62.1	57.6
StableVicuna-13B	38.5	37.6	55.9	52.8
Llama 2-13B	20.7	20.8	23.6	24
text-davinci-003	53.9	60.4	82.1	85.4

Table 1: Consistency comparison for all vs. accurate answers, for in-context or cross-temperature output generation.

Results

Figure 3 outlines a number of key findings for consistency and accuracy measurements without applying A2C. **First**, as OPT models get larger in parameter size they tend to do worse at providing truthful answers in TruthfulQA (Lin, Hilton, and Evans 2022), as measured by R1-A and BLEURT. Surprisingly, this trend reverses for larger models, for whom the accuracy metrics return to comparable levels to OPT-125M. **Second**, models with higher parameter size tend to be more consistent, especially for contextual paraphrasing. **Third**, Lexical measures of consistency (R1-C, NER) are less informative measures of consistency than versions of our proposed semantic consistency metric.

Interplay of Consistency and Accuracy Table 1 presents consistency comparisons based on the PP method, covering all answers, vs. only accurate answers. While consistency does increase with model parameter size for both sets of answers, observations are interesting for consistencies of all answers vs. correct answers for the same model. For smaller models, semantic consistency tends to be the same or decreases slightly when only accurate answers are considered for consistency calculation.

Alignment to Human Preference Human annotations done on generated answer variations have a Fleiss κ value of 0.9, indicating high inter-annotator agreement. Fig. 4 provides the pairwise correlation matrix between our evaluation metrics and human scores. Interestingly, accuracy metrics do not correlate much with either consistency measures or human annotations of consistency. The two accuracy measures show moderate positive correlation with one another ($\rho = 0.41$). Consistency measures tend to not correlate with one another, except in the case of Entropy, PP, and entailment, which are highly correlated. Entropy shows strong correlation with human annotations, followed by entailment, PP ($\rho = 0.83, 0.73, 0.55$, respectively). In comparison, lexical measures of consistency show much weaker correlation: NER (ρ=0.14) and R1-C (ρ=0.26). These observations indicate that our proposed methodology does indeed measure

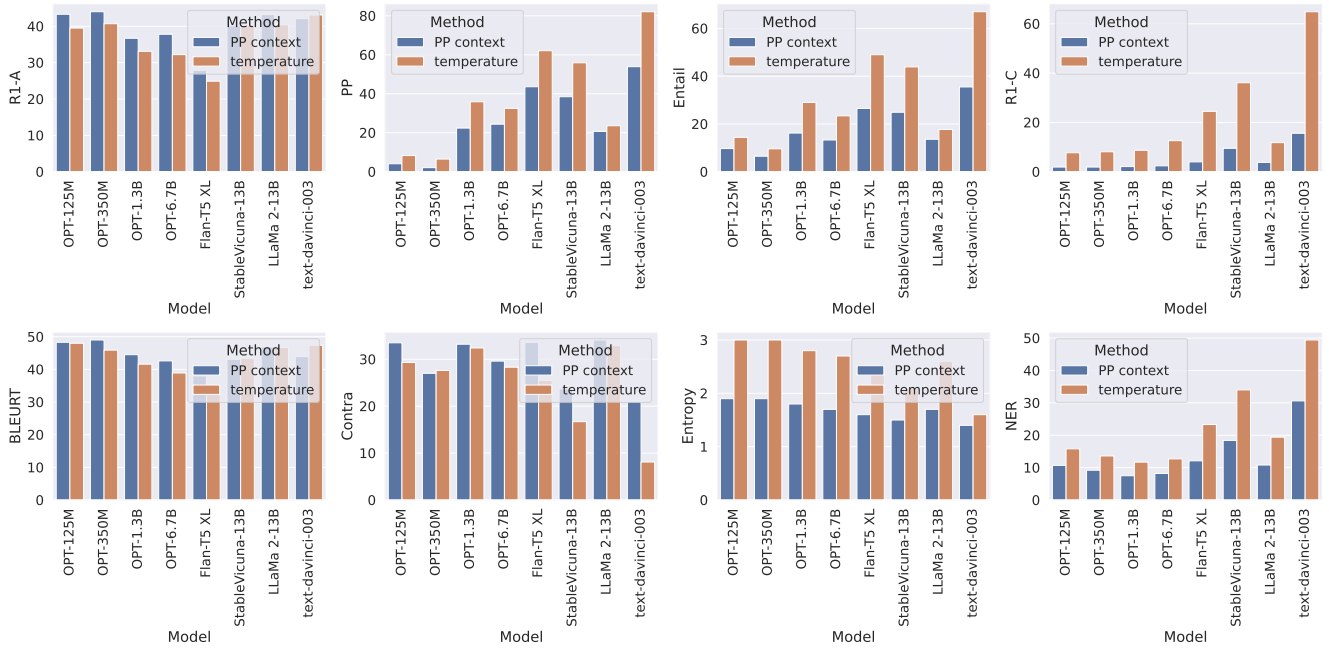

Figure 3: Accuracy and consistency of all models on TruthfulQA.

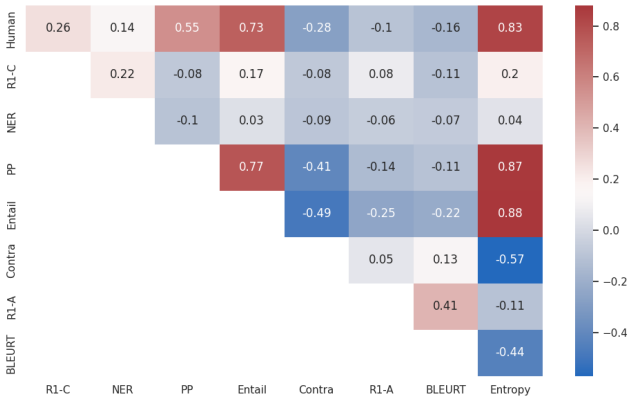

Figure 4: Pairwise correlations between all metrics and consistency annotations, for outputs from text-davinci-003 obtained using in-context paraphrasing.

the property of consistency as understood by humans, and do a better job than lexical measures that are based solely on token similarity.

Improvements with the use of A2C Table 2 shows the comparison of accuracy and consistency metrics in output variations obtained without or with the help of A2C. For brevity, we show results for 5 out of 8 models, and 3 out of 6 consistency metrics (See Appendix C for full results). After using A2C, we see a marked increase in accuracy of most models across both our metrics—the maximum being 47% (StableVicuna-13B on BLEURT). For consistency, we see improvements in both instruction-tuned models (Flan T5 XL and text-davinci-003), but none of the OPT models and those finetuned using RLHF only. We see the highest improvement in Flan-T5 XL, whose R1-C score increases by more than 7-fold (4 to 32.2).

Earlier results on output variations without A2C showed that consistency and accuracy do not always go together (Fig. 3 and Table 1). With the application of A2C, this difference is reconciled to an extent. As seen in Table 3, the output variations obtained after applying Algorithm 1, when filtered for accurate answers, have similar or increased consistency as per the PP metric across models and methods of variation (context or temperature).

Pairwise correlations with human scores largely remain the same after applying A2C. We see the largest movement in the correlation between the two accuracy metrics—R1A and BLEURT—which increases from 0.41 to 0.54. Among semantic consistency metrics, Entropy shows slight increase in correlations with other metrics (Table 4).

Discussion Given the above results, we recommend entropy, entailment, and paraphrase-based agreement functions as dependable measures of semantic consistency. Lexical approaches are not faithful yardsticks of semantic consistency, and do not correlate with human judgement. Many of our findings are intuitive or in line with previous results. For example, Lin, Hilton, and Evans (2022) found similar inverse scaling on TruthfulQA. Webson and Pavlick (2022) had shown that pretrained models have poor understanding of prompts, and prompts are better understood by larger models. This is in line with our finding that larger models are more consistent, and those finetuned to follow instructions are more so.

Less intuitive is the fact that consistency and accuracy do